Machine Learning

Machine Learning is a branch of Artificial Intelligence.

Types:

1. Supervised Learning
2. Unsupervised Learning
3. Reinforcement Learning

Applications:

- Healthcare
- Finance
- Recommendation Systems
- Computer Vision

Machine Learning Workflow

1. Data Collection

The input to the system is one or more PDF documents uploaded by the user. These documents may contain text, tables, reports, research papers, notes, or other textual information.

Purpose:

- Accept user-uploaded documents.
- Extract readable text.
- Prepare the document for further processing.

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2. Text Extraction

The uploaded PDF is processed using a PDF parsing library to extract textual content.

Techniques Used

- PDF Parsing
- Text Extraction
- Document Reading

Output

- Raw text from the uploaded document.

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3. Text Preprocessing

Before applying Machine Learning, the extracted text is cleaned and prepared.

Operations include

- Removing unnecessary spaces
- Eliminating empty lines
- Standardizing formatting
- Preserving meaningful document structure

Why preprocessing is important

- Improves embedding quality.
- Removes noise.
- Produces cleaner semantic representations.

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4. Text Chunking

Large documents cannot be processed efficiently as one continuous block. Therefore, the text is divided into smaller chunks.

Typical chunk size:

- 500–1000 characters

Overlap:

- 100–200 characters

Benefits

- Prevents information loss.
- Improves retrieval accuracy.
- Maintains context across chunk boundaries.
- Reduces computational complexity.

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5. Feature Extraction using Embeddings

Instead of representing text using traditional methods like Bag of Words or TF-IDF, the project uses modern embedding models.

An embedding model converts each text chunk into a high-dimensional numerical vector that captures semantic meaning.

Example:

"The capital of India is New Delhi."

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Embedding Vector

[0.12, -0.55, 0.91, ..., 0.34]

These vectors allow the system to identify semantically similar text even when different words are used.

Advantages

- Understands context.
- Captures semantic similarity.
- Handles synonyms effectively.
- Performs better than keyword-based search.

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6. Vector Database (FAISS)

After generating embeddings, they are stored in a FAISS vector database.

FAISS indexes the embedding vectors so that similar chunks can be retrieved quickly.

Functions

- Fast similarity search
- Efficient nearest-neighbor retrieval
- Scalable document indexing

Benefits

- Very low search latency
- Efficient memory usage
- Suitable for large document collections

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7. User Query Processing

When the user asks a question:

Example:

"What is the objective of this project?"

The system:

1. Converts the question into an embedding.
2. Searches FAISS for the most similar document chunks.
3. Retrieves the top relevant chunks.

This process is known as **semantic retrieval**.

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8. Retrieval-Augmented Generation (RAG)

Rather than answering from the LLM's general knowledge alone, the retrieved document chunks are supplied as context.

The language model then generates an answer grounded in the uploaded PDF.

Workflow:

User Question

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Embedding Generation

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Vector Search (FAISS)

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Relevant Chunks Retrieved

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LLM Receives Context

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Generated Answer

This reduces hallucinations and ensures responses are based on the uploaded document.

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9. Response Generation

The Large Language Model reads:

- User question
- Retrieved document chunks

It generates a fluent, context-aware, and document-based answer.

Characteristics:

- Human-like language
- Context-aware
- Accurate
- Easy to understand

Machine Learning Techniques Used

- Natural Language Processing (NLP)
- Semantic Embedding Generation
- Vector Similarity Search
- Retrieval-Augmented Generation (RAG)
- Context-Aware Text Generation
- Information Retrieval
- Document Chunking
- Dense Vector Representation

Why Embeddings Instead of Traditional Machine Learning?

Traditional approaches like TF-IDF or Bag of Words rely on exact word matches and struggle with synonyms or context.

Embedding models encode semantic meaning, allowing the system to retrieve relevant information even when the user's wording differs from the document.

Example:

Document:
"Artificial Intelligence improves healthcare."

User asks:
"How does AI help hospitals?"

A keyword search might miss this relationship, while embeddings recognize that the concepts are semantically related.

Advantages of the Machine Learning Approach

- Understands document context rather than keywords.
- Retrieves semantically relevant information.
- Produces accurate and context-based answers.
- Supports large PDF documents efficiently.
- Minimizes incorrect or hallucinated responses.

- Provides fast retrieval using vector indexing.
- Scales well to multiple uploaded documents.

Limitations

- Answer quality depends on the quality of the uploaded PDF.
- Scanned PDFs without OCR may not be processed correctly.
- Embedding generation can take time for very large documents.
- The LLM may still generate imperfect responses if the retrieved context is incomplete.
- Retrieval quality depends on chunk size, overlap, and embedding model selection.

Future Improvements

- OCR support for scanned PDFs.
- Multilingual document understanding.
- Hybrid retrieval combining semantic and keyword search.
- Automatic document summarization.
- Better reranking of retrieved chunks.
- Support for images, tables, and charts.
- Fine-tuned domain-specific embedding models.
- Multi-document comparison and citation support.

Conclusion

The Machine Learning pipeline in DocuMind AI combines NLP, embedding models, vector similarity search, and Retrieval-Augmented Generation to enable intelligent document question answering. By retrieving semantically relevant information from uploaded PDFs and providing it as context to a Large Language Model, the system delivers accurate, context-aware, and reliable responses while reducing hallucinations compared to traditional chatbot approaches.